

0707.4470: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0707.4470_E00025	17	■0.382	$\begin{aligned} & \backslash begin{aligned} & \& F=E_{\mathrm{x}}\,\mathrm{d}x\wedge\mathrm{d}t+E_{\mathrm{y}}\,\mathrm{d}y\wedge\mathrm{d}t+E_{\mathrm{z}}\,\mathrm{d}z\wedge\mathrm{d}t \\ & +B_{\mathrm{x}}\,\mathrm{d}y\wedge\mathrm{d}z+B_{\mathrm{y}}\,\mathrm{d}z\wedge\mathrm{d}x+B_{\mathrm{z}}\,\mathrm{d}x\wedge\mathrm{d}y, \\ & \backslash end{aligned} \end{aligned}$	$F = E_x \, dx \wedge dt + E_y \, dy \wedge dt + E_z \, dz \wedge dt \\ + B_x \, dy \wedge dz + B_y \, dz \wedge dx + B_z \, dx \wedge dy$	$F = E_x \, dx \wedge dt + E_y \, dy \wedge dt + E_z \, dz \wedge dt \\ + B_x \, dy \wedge dz + B_y \, dz \wedge dx + B_z \, dx \wedge dy,$
0707.4470_E00032	19	■0.582	$\begin{aligned} & \backslash begin{aligned} & \& \frac{\left(B_x\right)_{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1}-\left(B_x\right)_{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} \\ & \frac{\left(E_y\right)_{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}}-\left(E_y\right)_{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z}-\frac{\left(E_z\right)_{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}}-\left(E_z\right)_{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{\left(B_y\right)_{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1}-\left(B_y\right)_{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} \\ & \frac{\left(E_z\right)_{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}}-\left(E_z\right)_{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x}-\frac{\left(E_x\right)_{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}}-\left(E_x\right)_{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{\left(B_z\right)_{k+\frac{1}{2},l+\frac{1}{2},m}^{n+1}-\left(B_z\right)_{k+\frac{1}{2},l+\frac{1}{2},m}^n}{\Delta t} \\ & \frac{\left(E_x\right)_{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}}-\left(E_x\right)_{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y}-\frac{\left(E_y\right)_{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}}-\left(E_y\right)_{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x} \\ & \backslash end{aligned} \end{aligned}$	$\frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1}-B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y}$ $\frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1}-B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z}$ $\frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^{n+1}-B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^n}{\Delta t} =$ $\frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x}$	

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0707.4470_EQ0035	20	0.705	$\begin{aligned} & \frac{\left(D_x \right _{k+\frac{1}{2},l,m} - D_x \left _{k+\frac{1}{2},l,m}^{n-\frac{1}{2}} \right)}{\Delta t} = \\ & \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k+\frac{1}{2},l-\frac{1}{2},m}^n \right.}{\Delta y} - \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k+\frac{1}{2},l,m-\frac{1}{2}}^n \right.}{\Delta z} - J_x \left _{k+\frac{1}{2},l,m}^n \right. \\ & \frac{D_y \left _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} - D_y \left _{k,l+\frac{1}{2},m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l+\frac{1}{2},m-\frac{1}{2}}^n \right.}{\Delta z} - \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k-\frac{1}{2},l+\frac{1}{2},m}^n \right.}{\Delta x} - J_y \left _{k,l+\frac{1}{2},m}^n \right. \\ & \frac{D_z \left _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \left _{k,l,m+\frac{1}{2}}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k-\frac{1}{2},l,m+\frac{1}{2}}^n \right.}{\Delta x} - \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l-\frac{1}{2},m+\frac{1}{2}}^n \right.}{\Delta y} - J_z \left _{k,l,m+\frac{1}{2}}^n \right. \end{aligned}$	$\begin{aligned} & \frac{D_x \left _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}} - D_x \left _{k+\frac{1}{2},l,m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k+\frac{1}{2},l-\frac{1}{2},m}^n \right.}{\Delta y} - \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k+\frac{1}{2},l,m-\frac{1}{2}}^n \right.}{\Delta z} - J_x \left _{k+\frac{1}{2},l,m}^n \right. \\ & \frac{D_y \left _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} - D_y \left _{k,l+\frac{1}{2},m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l+\frac{1}{2},m-\frac{1}{2}}^n \right.}{\Delta z} - \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k-\frac{1}{2},l+\frac{1}{2},m}^n \right.}{\Delta x} - J_y \left _{k,l+\frac{1}{2},m}^n \right. \\ & \frac{D_z \left _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \left _{k,l,m+\frac{1}{2}}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k-\frac{1}{2},l,m+\frac{1}{2}}^n \right.}{\Delta x} - \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l-\frac{1}{2},m+\frac{1}{2}}^n \right.}{\Delta y} - J_z \left _{k,l,m+\frac{1}{2}}^n \right. \end{aligned}$	
0707.4470_EQ0020	12	0.761	$\int_{\partial \sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$	$\int_{\sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$	$\int_{\sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$
0707.4470_EQ0009	7	0.764	$S[A] = \int_X \mathcal{L}.$	$S[A] = \int_X \mathcal{L}.$	$S[A] = \int_X \mathcal{L}.$

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0707.4470_EQ0029	18	0.813	$\begin{aligned} & \text{\texttt{\begin{aligned} x z t \text{-face : } \& -\left(\right.} \\ & \left. \left. \left. E_{\left\{ x \right\}} \right _{\left\{ k+\frac{1}{2}, l, m+1 \right\}}^{\left\{ n+\frac{1}{2} \right\}} \right. \right. \\ & \left. \left. \left. E_{\left\{ x \right\}} \right _{\left\{ k+\frac{1}{2}, l, m \right\}}^{\left\{ n+\frac{1}{2} \right\}} \right. \right. \\ & \left. \left. \left. E_{\left\{ z \right\}} \right _{\left\{ k+1, l, m+\frac{1}{2} \right\}}^{\left\{ n+\frac{1}{2} \right\}} \right. \right. \\ & \left. \left. \left. E_{\left\{ z \right\}} \right _{\left\{ k, l, m+\frac{1}{2} \right\}}^{\left\{ n+\frac{1}{2} \right\}} \right. \right. \\ & \left. \left. \left. B_y \right _{\left\{ k+\frac{1}{2}, l, m+\frac{1}{2} \right\}}^{\left\{ n+1 \right\}} \right. \right. \\ & \left. \left. \left. B_y \right _{\left\{ k+\frac{1}{2}, l, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. \Delta x \Delta t \right. \right. \\ & \left. \left. \left. \Delta z \Delta t \right. \right. \\ & \left. \left. \left. \Delta x \Delta z \right. \right. \right. \end{aligned}}$	$\begin{aligned} xzt\text{-face : } & - \left(E_x \Big _{k+\frac{1}{2}, l, m+1}^{n+\frac{1}{2}} - E_x \Big _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \right) \Delta x \Delta t \\ & + \left(E_z \Big _{k+1, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z \Big _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \right) \Delta z \Delta t \\ & - \left(B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^{n+1} - B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \right) \Delta x \Delta z \end{aligned}$	$\begin{aligned} xzt\text{-face : } & - \left(E_x \Big _{k+\frac{1}{2}, l, m+1}^{n+\frac{1}{2}} - E_x \Big _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \right) \Delta x \Delta t \\ & + \left(E_z \Big _{k+1, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z \Big _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \right) \Delta z \Delta t \\ & - \left(B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^{n+1} - B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \right) \Delta x \Delta z \end{aligned}$
0707.4470_EQ0031	19	0.822	$\begin{aligned} & \text{\texttt{\begin{aligned} x y z \text{-face : } \& \left(\right.} \\ & \left. \left. \left. B_x \right _{\left\{ k+1, l+\frac{1}{2}, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. B_x \right _{\left\{ k, l+\frac{1}{2}, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. B_y \right _{\left\{ k+\frac{1}{2}, l+1, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. B_y \right _{\left\{ k+\frac{1}{2}, l, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. B_z \right _{\left\{ k+\frac{1}{2}, l+\frac{1}{2}, m+1 \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. B_z \right _{\left\{ k+\frac{1}{2}, l+\frac{1}{2}, m \right\}}^{\left\{ n \right\}} \right. \right. \\ & \left. \left. \left. \Delta y \Delta z \right. \right. \\ & \left. \left. \left. \Delta x \Delta z \right. \right. \\ & \left. \left. \left. \Delta x \Delta y \right. \right. \right. \end{aligned}}$	$\begin{aligned} xyz\text{-face : } & \left(B_x \Big _{k+1, l+\frac{1}{2}, m+\frac{1}{2}}^n - B_x \Big _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \right) \Delta y \Delta z \\ & + \left(B_y \Big _{k+\frac{1}{2}, l+1, m+\frac{1}{2}}^n - B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \right) \Delta x \Delta z \\ & + \left(B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m+1}^n - B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \right) \Delta x \Delta y \end{aligned}$	$\begin{aligned} xyz\text{-face : } & \left(B_x \Big _{k+1, l+\frac{1}{2}, m+\frac{1}{2}}^n - B_x \Big _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \right) \Delta y \Delta z \\ & + \left(B_y \Big _{k+\frac{1}{2}, l+1, m+\frac{1}{2}}^n - B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \right) \Delta x \Delta z \\ & + \left(B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m+1}^n - B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \right) \Delta x \Delta y \end{aligned}$
0707.4470_EQ0026	18	0.856	$\begin{aligned} & \text{\texttt{\begin{gathered} x t \text{-face : } \backslash\left. E_{\left\{ x \right\}} \right _{\left\{ k+\frac{1}{2}, l, m \right\}}^{\left\{ n+\frac{1}{2} \right\}} \Delta x \Delta t} \\ y t \text{-face : } \backslash\left. E_{\left\{ y \right\}} \right _{\left\{ k, l+\frac{1}{2}, m \right\}}^{\left\{ n+\frac{1}{2} \right\}} \Delta y \Delta t} \\ z t \text{-face : } \backslash\left. E_{\left\{ z \right\}} \right _{\left\{ k, l, m+\frac{1}{2} \right\}}^{\left\{ n+\frac{1}{2} \right\}} \Delta z \Delta t} \\ y z \text{-face : } B_x \Big _{\left\{ k, l+\frac{1}{2}, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \Delta y \Delta z} \\ x z \text{-face : } B_y \Big _{\left\{ k+\frac{1}{2}, l, m+\frac{1}{2} \right\}}^{\left\{ n \right\}} \Delta z \Delta x} \\ x y \text{-face : } B_z \Big _{\left\{ k+\frac{1}{2}, l+\frac{1}{2}, m \right\}}^{\left\{ n \right\}} \Delta x \Delta y} \end{gathered}}$	$\begin{aligned} xt\text{-face : } & E_x \Big _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ yt\text{-face : } & E_y \Big _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ zt\text{-face : } & E_z \Big _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \Delta z \Delta t \\ yz\text{-face : } & B_x \Big _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \Delta y \Delta z \\ xz\text{-face : } & B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \Delta z \Delta x \\ xy\text{-face : } & B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \Delta x \Delta y \end{aligned}$	$\begin{aligned} xt\text{-face : } & E_x \Big _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ yt\text{-face : } & E_y \Big _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ zt\text{-face : } & E_z \Big _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \Delta z \Delta t \\ yz\text{-face : } & B_x \Big _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \Delta y \Delta z \\ xz\text{-face : } & B_y \Big _{k+\frac{1}{2}, l, m+\frac{1}{2}}^n \Delta z \Delta x \\ xy\text{-face : } & B_z \Big _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \Delta x \Delta y. \end{aligned}$

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0707.4470_E00030	18	■ 0.881	$\begin{aligned} & y \, z \, t \text{-face: } \& -\left(\left. E_y \right _{k, l+\frac{1}{2}, m+1}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. E_z \right _{k, l+1, m+\frac{1}{2}}^{n+\frac{1}{2}} - \left. E_z \right _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \right) \Delta z \Delta t \\ & + \left(\left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^{n+1} - \left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \right) \Delta y \Delta z \end{aligned}$	$\begin{aligned} & yzt\text{-face: } - \left(\left. E_y \right _{k, l+\frac{1}{2}, m+1}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. E_z \right _{k, l+1, m+\frac{1}{2}}^{n+\frac{1}{2}} - \left. E_z \right _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \right) \Delta z \Delta t \\ & + \left(\left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^{n+1} - \left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \right) \Delta y \Delta z \end{aligned}$	$\begin{aligned} yzt\text{-face: } & - \left(\left. E_y \right _{k, l+\frac{1}{2}, m+1}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. E_z \right _{k, l+1, m+\frac{1}{2}}^{n+\frac{1}{2}} - \left. E_z \right _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \right) \Delta z \Delta t \\ & + \left(\left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^{n+1} - \left. B_x \right _{k, l+\frac{1}{2}, m+\frac{1}{2}}^n \right) \Delta y \Delta z \end{aligned}$
0707.4470_E00058 (5.3)	32	■ 0.887	$\begin{aligned} & \mathbf{d} S[A] \cdot \alpha = \int_X (\mathrm{d}_t \alpha \wedge D - \mathrm{d}_s \alpha \wedge H \wedge \mathrm{d} t + \alpha \wedge J \wedge \mathrm{d} t) \\ & = \int_X \alpha \wedge (\mathrm{d}_t D - \mathrm{d}_s H \wedge \mathrm{d} t + J \wedge \mathrm{d} t) \end{aligned}$	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha &= \int_X (\mathrm{d}_t \alpha \wedge D - \mathrm{d}_s \alpha \wedge H \wedge \mathrm{d} t + \alpha \wedge J \wedge \mathrm{d} t) \\ &= \int_X \alpha \wedge (\mathrm{d}_t D - \mathrm{d}_s H \wedge \mathrm{d} t + J \wedge \mathrm{d} t). \end{aligned}$	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha &= \int_X (\mathrm{d}_t \alpha \wedge D - \mathrm{d}_s \alpha \wedge H \wedge \mathrm{d} t + \alpha \wedge J \wedge \mathrm{d} t) \\ &= \int_X \alpha \wedge (\mathrm{d}_t D - \mathrm{d}_s H \wedge \mathrm{d} t + J \wedge \mathrm{d} t). \end{aligned}$
0707.4470_E00059	32	■ 0.892	$\mathbf{d} S[A] \cdot \alpha = \int_X \alpha \wedge D \Big _{t_0}^{t_f},$	$\mathbf{d} S[A] \cdot \alpha = \int_X \alpha \wedge D \Big _{t_0}^{t_f},$	$\mathbf{d} S[A] \cdot \alpha = \int_X \alpha \wedge D \Big _{t_0}^{t_f},$
0707.4470_E00016 (2.5)	8	■ 0.911	$\begin{aligned} & \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) = \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathrm{d} \alpha) \wedge G + (F - \mathrm{d} A) \wedge \gamma] \\ & = \int_X [\alpha \wedge (\mathcal{J} - \mathrm{d} G) + \phi \wedge (G - *F) + (F - \mathrm{d} A) \wedge \gamma] \end{aligned}$	$\begin{aligned} \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) &= \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathrm{d} \alpha) \wedge G + (F - \mathrm{d} A) \wedge \gamma] \\ &= \int_X [\alpha \wedge (\mathcal{J} - \mathrm{d} G) + \phi \wedge (G - *F) + (F - \mathrm{d} A) \wedge \gamma] \end{aligned}$	$\begin{aligned} \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) &= \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathrm{d} \alpha) \wedge G + (F - \mathrm{d} A) \wedge \gamma] \\ &= \int_X [\alpha \wedge (\mathcal{J} - \mathrm{d} G) + \phi \wedge (G - *F) + (F - \mathrm{d} A) \wedge \gamma] \end{aligned}$
0707.4470_E00028	18	■ 0.937	$\begin{aligned} & x \, y \, t \text{-face: } \& - \left(\left. E_x \right _{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}} - \left. E_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \right) \Delta x \Delta t \\ & + \left(\left. E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - \left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \right) \Delta x \Delta y \end{aligned}$	$\begin{aligned} xyt\text{-face: } & - \left(\left. E_x \right _{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}} - \left. E_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \right) \Delta x \Delta t \\ & + \left(\left. E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - \left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \right) \Delta x \Delta y \end{aligned}$	$\begin{aligned} xyt\text{-face: } & - \left(\left. E_x \right _{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}} - \left. E_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \right) \Delta x \Delta t \\ & + \left(\left. E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - \left. E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \right) \Delta y \Delta t \\ & + \left(\left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - \left. B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \right) \Delta x \Delta y \end{aligned}$

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0707.4470_ E00036 (4.2)	20	0.951	$\begin{aligned} & \frac{\left. D_x \right _{k+\frac{1}{2},l,m} - \left. D_x \right _{k-\frac{1}{2},l,m}}{\Delta x} + \frac{\left. D_y \right _{k,l+\frac{1}{2},m} - \left. D_y \right _{k,l-\frac{1}{2},m}}{\Delta y} \\ & + \frac{\left. D_z \right _{k,l,m+\frac{1}{2}} - \left. D_z \right _{k,l,m-\frac{1}{2}}}{\Delta z} = \rho _{k,l,m} . \end{aligned}$	$\frac{D_x _{k+\frac{1}{2},l,m} - D_x _{k-\frac{1}{2},l,m}}{\Delta x} + \frac{D_y _{k,l+\frac{1}{2},m} - D_y _{k,l-\frac{1}{2},m}}{\Delta y} + \frac{D_z _{k,l,m+\frac{1}{2}} - D_z _{k,l,m-\frac{1}{2}}}{\Delta z} = \rho _{k,l,m} .$	$\frac{D_x _{k+\frac{1}{2},l,m} - D_x _{k-\frac{1}{2},l,m}}{\Delta x} + \frac{D_y _{k,l+\frac{1}{2},m} - D_y _{k,l-\frac{1}{2},m}}{\Delta y} + \frac{D_z _{k,l,m+\frac{1}{2}} - D_z _{k,l,m-\frac{1}{2}}}{\Delta z} = \rho _{k,l,m} .$
0707.4470_E00023	13	0.955	$\begin{aligned} (\alpha, \beta) &= \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \\ &= \sum_{\sigma^k} \kappa(\sigma) \frac{ * \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \end{aligned}$	$(\alpha, \beta) = \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle = \sum_{\sigma^k} \kappa(\sigma) \frac{ * \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle$	$\begin{aligned} (\alpha, \beta) &= \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \\ &= \sum_{\sigma^k} \kappa(\sigma) \frac{ * \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \end{aligned}$
0707.4470_E00008	7	0.963	$\mathrm{d} \mathscr{L} = - \frac{1}{2} \, \mathrm{d} A \wedge * \, \mathrm{d} A + A \wedge \mathscr{J} ,$	$\mathcal{L} = -\frac{1}{2} \, \mathrm{d} A \wedge * \, \mathrm{d} A + A \wedge \mathscr{J} ,$	$\mathcal{L} = -\frac{1}{2} \mathrm{d} A \wedge * \mathrm{d} A + A \wedge \mathscr{J} ,$
0707.4470_ E00012 (2.3)	8	0.982	$\mathrm{d} F = 0$ $\mathrm{d} G = \mathscr{J}$	$\mathrm{d} F = 0$ $\mathrm{d} G = \mathscr{J}$	$\mathrm{d} F = 0$ $\mathrm{d} G = \mathscr{J}$
0707.4470_E00060	32	0.982	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$
0707.4470_ E00046 (4.4)	24	0.988	$\frac{D_e^{m+1/2} - D_e^{m-1/2}}{t_e^{m+1/2} - t_e^{m-1/2}} = \mathrm{d}_1^T \left(H_{\sigma}^n \mathbb{K} \{ t_{\sigma}^n = t_e^m \} \right) - J_e^m ,$	$\frac{D_e^{m+1/2} - D_e^{m-1/2}}{t_e^{m+1/2} - t_e^{m-1/2}} = \mathrm{d}_1^T \left(H_{\sigma}^n \mathbb{K} \{ t_{\sigma}^n = t_e^m \} \right) - J_e^m ,$	$\frac{D_e^{m+1/2} - D_e^{m-1/2}}{t_e^{m+1/2} - t_e^{m-1/2}} = \mathrm{d}_1^T \left(H_{\sigma}^n \mathbb{K} \{ t_{\sigma}^n = t_e^m \} \right) - J_e^m ,$
0707.4470_E00002	6	0.996	$\oint_C \mathbf{E} \cdot \mathrm{d} \mathbf{l} = - \frac{\mathrm{d}}{\mathrm{d} t} \int_S \mathbf{B} \cdot \mathrm{d} \mathbf{A} ,$	$\oint_C \mathbf{E} \cdot \mathrm{d} \mathbf{l} = - \frac{\mathrm{d}}{\mathrm{d} t} \int_S \mathbf{B} \cdot \mathrm{d} \mathbf{A} ,$	$\oint_C \mathbf{E} \cdot \mathrm{d} \mathbf{l} = - \frac{\mathrm{d}}{\mathrm{d} t} \int_S \mathbf{B} \cdot \mathrm{d} \mathbf{A} ,$
0707.4470_ E00013 (2.4)	8	0.996	$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$ $\nabla \cdot \mathbf{B} = 0 .$	$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$ $\nabla \cdot \mathbf{B} = 0 .$	$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$ $\nabla \cdot \mathbf{B} = 0 .$

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0707.4470_ EQ0011 (2.1)	7	■ 1.000	$\mathrm{d} \ast \mathrm{d} A = \mathscr{J} \text{ .}$	$\mathrm{d} \ast \mathrm{d} A = \mathscr{J} \text{ .}$	$\mathrm{d} \ast \mathrm{d} A = \mathscr{J} \text{ .}$
0707.4470_ EQ0014 (2.6)	8	■ 1.000	$\begin{aligned} \nabla \times \mathbf{H} - \partial_t \mathbf{D} &= \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho \text{ .} \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{H} - \partial_t \mathbf{D} &= \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho \text{ .} \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{H} - \partial_t \mathbf{D} &= \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho \text{ .} \end{aligned}$
0707.4470_ EQ0017 (2.2)	8	■ 1.000	$\mathrm{d} G = \mathscr{J} \text{ , } \quad G = \ast F \text{ , } \quad F = \mathrm{d} A \text{ .}$	$\mathrm{d} G = \mathscr{J} \text{ , } \quad G = \ast F \text{ , } \quad F = \mathrm{d} A \text{ .}$	$\mathrm{d} G = \mathscr{J} \text{ , } \quad G = \ast F \text{ , } \quad F = \mathrm{d} A \text{ .}$
0707.4470_EQ0018	9	■ 1.000	$\partial_t (\nabla \cdot \mathbf{B}) = 0 \text{ , } \quad \partial_t (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{J} = \partial_t (\nabla \cdot \mathbf{D} - \rho) = 0 \text{ .}$	$\partial_t (\nabla \cdot \mathbf{B}) = 0 \text{ , } \quad \partial_t (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{J} = \partial_t (\nabla \cdot \mathbf{D} - \rho) = 0 \text{ .}$	$\partial_t (\nabla \cdot \mathbf{B}) = 0 \text{ , } \quad \partial_t (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{J} = \partial_t (\nabla \cdot \mathbf{D} - \rho) = 0 \text{ .}$
0707.4470_EQ0019	12	■ 1.000	$\langle \alpha, \sigma \rangle \equiv \int_{\sigma} \alpha \text{ .}$	$\langle \alpha, \sigma \rangle \equiv \int_{\sigma} \alpha \text{ .}$	$\langle \alpha, \sigma \rangle \equiv \int_{\sigma} \alpha \text{ .}$
0707.4470_EQ0021	12	■ 1.000	$\langle \mathrm{d} \alpha, \sigma \rangle = \langle \alpha, \partial \sigma \rangle \text{ ,}$	$\langle \mathrm{d} \alpha, \sigma \rangle = \langle \alpha, \partial \sigma \rangle \text{ ,}$	$\langle \mathrm{d} \alpha, \sigma \rangle = \langle \alpha, \partial \sigma \rangle \text{ ,}$
0707.4470_EQ0022	13	■ 1.000	$\frac{1}{ \ast \sigma } \langle \ast \alpha, \ast \sigma \rangle = \kappa(\sigma) \frac{1}{ \sigma } \langle \alpha, \sigma \rangle \text{ ,}$	$\frac{1}{ \ast \sigma } \langle \ast \alpha, \ast \sigma \rangle = \kappa(\sigma) \frac{1}{ \sigma } \langle \alpha, \sigma \rangle \text{ ,}$	$\frac{1}{ \ast \sigma } \langle \ast \alpha, \ast \sigma \rangle = \kappa(\sigma) \frac{1}{ \sigma } \langle \alpha, \sigma \rangle \text{ ,}$
0707.4470_ EQ0024 (3.1)	16	■ 1.000	$(\mathrm{d} \alpha, \beta) = (\alpha, \delta \beta) + \langle \alpha \wedge \ast \beta, \partial K \rangle \text{ .}$	$(\mathrm{d} \alpha, \beta) = (\alpha, \delta \beta) + \langle \alpha \wedge \ast \beta, \partial K \rangle \text{ .}$	$(\mathrm{d} \alpha, \beta) = (\alpha, \delta \beta) + \langle \alpha \wedge \ast \beta, \partial K \rangle \text{ .}$
0707.4470_EQ0027	18	■ 1.000	$\mathrm{d} F = 0 \text{ , } \quad \mathrm{d} G = \mathscr{J} \text{ ,}$	$\mathrm{d} F = 0 \text{ , } \quad \mathrm{d} G = \mathscr{J} \text{ ,}$	$\mathrm{d} F = 0 \text{ , } \quad \mathrm{d} G = \mathscr{J} \text{ ,}$
0707.4470_EQ0034	19	■ 1.000	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E} \text{ , } \quad \nabla \cdot \mathbf{B} = 0 \text{ .}$	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E} \text{ , } \quad \nabla \cdot \mathbf{B} = 0 \text{ .}$	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E} \text{ , } \quad \nabla \cdot \mathbf{B} = 0 \text{ .}$
0707.4470_EQ0037	20	■ 1.000	$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J} \text{ , } \quad \nabla \cdot \mathbf{D} = \rho \text{ .}$	$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J} \text{ , } \quad \nabla \cdot \mathbf{D} = \rho \text{ .}$	$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J} \text{ , } \quad \nabla \cdot \mathbf{D} = \rho \text{ .}$
0707.4470_EQ0038	21	■ 1.000	$F = E \wedge \mathrm{d} t + B \text{ .}$	$F = E \wedge \mathrm{d} t + B \text{ .}$	$F = E \wedge \mathrm{d} t + B \text{ .}$
0707.4470_EQ0039	22	■ 1.000	$\frac{B^{n+1} - B^n}{\Delta t} = -\mathrm{d}_1 E^{n+1/2} \text{ .}$	$\frac{B^{n+1} - B^n}{\Delta t} = -\mathrm{d}_1 E^{n+1/2} \text{ .}$	$\frac{B^{n+1} - B^n}{\Delta t} = -\mathrm{d}_1 E^{n+1/2} \text{ .}$
0707.4470_EQ0040	22	■ 1.000	$\frac{D^{n+1/2} - D^{n-1/2}}{\Delta t} = \mathrm{d}_1^T H^n - J^n \text{ .}$	$\frac{D^{n+1/2} - D^{n-1/2}}{\Delta t} = \mathrm{d}_1^T H^n - J^n \text{ .}$	$\frac{D^{n+1/2} - D^{n-1/2}}{\Delta t} = \mathrm{d}_1^T H^n - J^n \text{ .}$
0707.4470_EQ0041	23	■ 1.000	$\Theta_{\sigma} = \{t_{\sigma}^0 < \dots < t_{\sigma}^{N_{\sigma}}\} \text{ .}$	$\Theta_{\sigma} = \{t_{\sigma}^0 < \dots < t_{\sigma}^{N_{\sigma}}\} \text{ .}$	$\Theta_{\sigma} = \{t_{\sigma}^0 < \dots < t_{\sigma}^{N_{\sigma}}\} \text{ .}$
0707.4470_EQ0042	23	■ 1.000	$\Theta_e = \bigcup_{\sigma \ni e} \Theta_{\sigma} = \{t_e^0 \leq \dots \leq t_e^{N_e}\} \text{ .}$	$\Theta_e = \bigcup_{\sigma \ni e} \Theta_{\sigma} = \{t_e^0 \leq \dots \leq t_e^{N_e}\} \text{ .}$	$\Theta_e = \bigcup_{\sigma \ni e} \Theta_{\sigma} = \{t_e^0 \leq \dots \leq t_e^{N_e}\} \text{ .}$
0707.4470_EQ0043	23	■ 1.000	$\Theta_e' = \{t_e^{1/2} \leq \dots \leq t_e^{N_e-1/2}\} \text{ ,}$	$\Theta_e' = \{t_e^{1/2} \leq \dots \leq t_e^{N_e-1/2}\} \text{ ,}$	$\Theta_e' = \{t_e^{1/2} \leq \dots \leq t_e^{N_e-1/2}\} \text{ ,}$
0707.4470_EQ0044	24	■ 1.000	$\Theta_{\ast \sigma} = \Theta_{\sigma} \text{ , } \quad \Theta_{\ast e} = \Theta_e \text{ .}$	$\Theta_{\ast \sigma} = \Theta_{\sigma} \text{ , } \quad \Theta_{\ast e} = \Theta_e \text{ .}$	$\Theta_{\ast \sigma} = \Theta_{\sigma} \text{ , } \quad \Theta_{\ast e} = \Theta_e \text{ .}$
0707.4470_ EQ0045 (4.3)	24	■ 1.000	$\frac{B_{\sigma}^{n+1} - B_{\sigma}^n}{t_{\sigma}^{n+1} - t_{\sigma}^n} = -\mathrm{d}_1 \sum \left\{ E_e^{m+1/2} : t_{\sigma}^n < t_e^{m+1/2} < t_{\sigma}^{n+1} \right\} \text{ .}$	$\frac{B_{\sigma}^{n+1} - B_{\sigma}^n}{t_{\sigma}^{n+1} - t_{\sigma}^n} = -\mathrm{d}_1 \sum \left\{ E_e^{m+1/2} : t_{\sigma}^n < t_e^{m+1/2} < t_{\sigma}^{n+1} \right\} \text{ .}$	$\frac{B_{\sigma}^{n+1} - B_{\sigma}^n}{t_{\sigma}^{n+1} - t_{\sigma}^n} = -\mathrm{d}_1 \sum \left\{ E_e^{m+1/2} : t_{\sigma}^n < t_e^{m+1/2} < t_{\sigma}^{n+1} \right\} \text{ .}$
0707.4470_EQ0048	29	■ 1.000	$S_d[A] = \langle \mathscr{L}_d, K \rangle \text{ .}$	$S_d[A] = \langle \mathscr{L}_d, K \rangle \text{ .}$	$S_d[A] = \langle \mathscr{L}_d, K \rangle \text{ .}$
0707.4470_ EQ0051 (5.1)	30	■ 1.000	$\mathrm{d} S_d(A) \cdot \alpha = \langle \alpha \wedge \ast \mathrm{d} A, \partial K \rangle$	$\mathrm{d} S_d(A) \cdot \alpha = \langle \alpha \wedge \ast \mathrm{d} A, \partial K \rangle$	$\mathrm{d} S_d(A) \cdot \alpha = \langle \alpha \wedge \ast \mathrm{d} A, \partial K \rangle$

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0707.4470_EQ0052	30	■ 1.000	$\theta_{\mathscr{L}} \cdot \alpha = \alpha \wedge * \mathrm{d} A$.	$\theta_{\mathscr{L}_d} \cdot \alpha = \alpha \wedge * \mathrm{d} A$.	$\theta_{\mathscr{L}_d} \cdot \alpha = \alpha \wedge * \mathrm{d} A$.
0707.4470_EQ0053	30	■ 1.000	$\mathbf{d}^2 S_d[A] \cdot \alpha \cdot \beta = \langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle$.	$\mathbf{d}^2 S_d[A] \cdot \alpha \cdot \beta = \langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle$.	$\mathbf{d}^2 S_d[A] \cdot \alpha \cdot \beta = \langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle$.
0707.4470_EQ0054 (5.2)	30	■ 1.000	$\left\langle \omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta, \partial K \right\rangle = 0$	$\left\langle \omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta, \partial K \right\rangle = 0$	$\left\langle \omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta, \partial K \right\rangle = 0$
0707.4470_EQ0055	31	■ 1.000	$A = A_x \mathrm{d} x + A_y \mathrm{d} y + A_z \mathrm{d} z$.	$A = A_x \mathrm{d} x + A_y \mathrm{d} y + A_z \mathrm{d} z$.	$A = A_x \mathrm{d} x + A_y \mathrm{d} y + A_z \mathrm{d} z$.
0707.4470_EQ0056	32	■ 1.000	$\mathrm{d}_t A = E \wedge \mathrm{d} t, \quad \mathrm{d}_s A = B$.	$\mathrm{d}_t A = E \wedge \mathrm{d} t, \quad \mathrm{d}_s A = B$.	$\mathrm{d}_t A = E \wedge \mathrm{d} t, \quad \mathrm{d}_s A = B$.
0707.4470_EQ0057	32	■ 1.000	$\begin{aligned} \mathscr{L} &= -\frac{1}{2} (\mathrm{d}_t A + \mathrm{d}_s A) \wedge * (\mathrm{d}_t A + \mathrm{d}_s A) + A \wedge \mathscr{J} \\ &= -\frac{1}{2} (\mathrm{d}_t A \wedge * \mathrm{d}_t A + \mathrm{d}_s A \wedge * \mathrm{d}_s A) + A \wedge J \wedge \mathrm{d} t \end{aligned}$	$\begin{aligned} \mathscr{L} &= -\frac{1}{2} (\mathrm{d}_t A + \mathrm{d}_s A) \wedge * (\mathrm{d}_t A + \mathrm{d}_s A) + A \wedge \mathscr{J} \\ &= -\frac{1}{2} (\mathrm{d}_t A \wedge * \mathrm{d}_t A + \mathrm{d}_s A \wedge * \mathrm{d}_s A) + A \wedge J \wedge \mathrm{d} t \end{aligned}$	$\begin{aligned} \mathscr{L} &= -\frac{1}{2} (\mathrm{d}_t A + \mathrm{d}_s A) \wedge * (\mathrm{d}_t A + \mathrm{d}_s A) + A \wedge \mathscr{J} \\ &= -\frac{1}{2} (\mathrm{d}_t A \wedge * \mathrm{d}_t A + \mathrm{d}_s A \wedge * \mathrm{d}_s A) + A \wedge J \wedge \mathrm{d} t \end{aligned}$
0707.4470_EQ0061	32	■ 1.000	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_{\Sigma} f \wedge \rho \Big _{t_0}^{t_f}$.	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_{\Sigma} f \wedge \rho \Big _{t_0}^{t_f}$.	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_{\Sigma} f \wedge \rho \Big _{t_0}^{t_f}$.
0707.4470_EQ0062	33	■ 1.000	$(\mathrm{d}_s D - \rho) \Big _{t_0}^{t_f} = 0$.	$(\mathrm{d}_s D - \rho) \Big _{t_0}^{t_f} = 0$.	$(\mathrm{d}_s D - \rho) \Big _{t_0}^{t_f} = 0$.